

Quote No OF13004050/5

Date 28.06.2025

SHORT QUOTATION / OFFER



MIKRON HSM 500 - Heidenhain TNC 640

CUSTOMER INFORMATION

엠앤씨전자 비나

M&C Electronics Vina

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Product MIKRON HSM 500 - Heidenhain TNC 640

EXECUTIVE SUMMARY

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MIKRON HSM 500 - Heidenhain TNC 640



Item	Description	Quantity
MACHINE		
995001123	MIKRON HSM 500 - Heidenhain TNC 640	1 PC
600.50/01	Machine with EC conformity declaration	1 PC
205.10/01	Machine encl. polycarb. window	1 PC
205.08/03	Spindle 42000 min ⁻¹ / HSK-E40	1 PC
205.09/03	Tool magazine 36× HSK-E40	1 PC
600.12/58	Robot interface	1 PC
CHIP DISPOSITION		
205.11/01	Coolant tank	1 PC
COOLANT & DUST EXTRACTION		
292.12/13	Mist extraction unit	1 PC
292.12/20	Air blow off external	1 PC
OPTIONS		
600.12/19	Beacon	1 PC
PRECISION & ACCURACY		
600.12/13	Laser measuring system	1 PC
600.12/20	Tool blow-off unit	1 PC

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Item	Description	Quantity
600.12/08	Touch probe infrared HSK-E40	1 PC
600.12/10	Preparation touch probe infrared	1 PC

SMART MACHINE

500.52/01	APS Advanced Process System	1 PC
500.52/07	ITC Intelligent Thermal Control	1 PC
500.52/19	OSS Operator Support System	1 PC
500.52/13	DNC Remote network access on control	1 PC

PACKING & DOCUMENTATION

600.21/02	Packing HSM 500 Type C	1 PC
600.21/80	Preparation packing Type C	1 PC

Contract type embedded with the machine

CSP995011	Bronze - Recover fast	1 PC
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Other service contracts

200011529	My rConnect - Remote Access - Trial 1st year	1 PC
200011531	My rConnect - Live Data - Trial 1st year	1 PC

THIRD PARTY PRODUCTS

Z100002	Air Dryer	1 PC
Z100002	Stabilizer	1 PC

SERVICES

Z100003	Basic machine Installation	1 PC
Z100003	Operation Training	1 PC

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CONFIGURATION TOTAL

OFFER PRICE

USD 285.000

Freight charges are quoted on the basis of current estimates. Additional charges may apply in case market prices change significantly until actual transport occurs.

Quotation valid to 23.07.2025

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TERMS AND CONDITIONS

Delivery

계약 후 5~6 개월 이내

본 계약에서 납기라 함은 계약 목적물을 "매수인"이 정하는 장소에 "매도인"이 인도하는 시점을 말한다. 단, 원자재 공급, 노사분규, 차량 운송 사고, 정부의 조치 등 "매도인"의 납품 지연 사유와 납품 장소의 완공 지연, 부대 시설 미비, 노사분규, 천재지변에 의한 "매수인"의 인수 지연 사유가 발생할 시 상호 협의하여 인도 시기를 연장할 수 있다.

"매수인"은 지연이 "매도인"의 잘못으로 인해 발생했으며 "매수인"이 그러한 지연으로 인해 손실을 입었다는 사실이 입증될 수 있는 한 지연된 배송에 대해 청산된 손해배상을 청구할 수 있다. 다만 고객에게 대체 자재를 공급할 수 있는 경우, 후자는 지연에 대한 손해배상을 청구할 수 없다. 배송 지연에 대한 손해배상은 매주 지연될 때마다 0.5%를 초과해서는 안 되며, 어떤 경우에도 지연된 공급물품 계약 가격의 5%를 초과할 수 없다. 지연의 첫 2주 동안은 어떠한 손해배상도 지급되지 않는다. 지연은 이 조항에 명시적으로 규정된 권리와 청구권 외에 고객에게 어떠한 권리와 청구권도 부여하지 않는다.

Prices

본선 인도 조건

계약물품의 인도방법은 "매수인"이 지정한 선박에 "매도인"이 약속한 화물을 적재, 본선상에서 화물의 인도를 마칠때까지로 한다.

Payment

계약금: 계약시 30%

중도금: 납품시 60%

잔금: FAC후 10%

"매수인"은 본 계약물의 인수 후 대금 지불 조건대로 대금 지불을 이행하여야 한다. 대금의 지불 수단이 어음인 경우, 그에 상응하는 담보를 "매도인"에게 제공한다. "매수인"의 대금 지불 및 수금에 대한 제반사항은 "매도인"이 수행하며, 은행자금 및 리스 자금으로 지불시에는 반드시 "매도인"에게 지불한다. "매수인"이 본 계약의 대금 지불, 일자 및 지불 조건을 이행하지 못하였을 경우, "매도인"은 "매수인"에게 연리 5%의 연체 이자를 청구할 수 있다.

무역 관세 정책의 불확실성과 관련된 GF Machine Solutions의 판매 조건 변경

"판매자는 정부의 조치(수입/수출 관련 비용, 관세, 기타 제세공과금의 변경을 포함하되 이에 한정되지 아니함)로 인해 견적상 상품 가격에 영향을 미치는 사정이 발생한 경우, 선적 이전에 해당 가격을 조정할 수 있는 권리를 보유한다."

Warranty

설치 후 12개월

본 계약물의 품질보증기간은 "매도인"이 계약 물품을 "매수인"에게 인도한 날로부터 12개월 동안이나 이 기간중 "매수인"이 과실이나 고의에 의한 하자 발생시 상호 협의하여 유상처리한다.

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Reservation of title

1항 : "매수인"이 본 계약물에 대하여 대금 지불 조건을 이행하지 않을 경우 "매도인"의 서면 통지로 계약은 해지되며 "매수인"은 즉시 계약 목적물을 "매도인"에게 반환한다.

2항 : "매수인"은 본 계약물의 대금 완제시까지 "매도인"의 승인 없이는 물품을 제3자의 채무 변제를 하기 위한 저당 설정 등 어떠한 형태로의 처분도 할 수 없다.

3항 : "매수인"이 주소를 이전하였거나 "매도인"의 채권 보전 등 계약 이행에 영향을 미치는 사유가 발생하거나 발생할 우려가 있는 경우 "매수인"은 이를 즉시 "매도인"에게 통보하여야 한다.

Limitation of Liability

계약 또는 그 위반과 관련하여 "매수인"의 클레임이 존재 하는 경우, 청구 총액은 각 구매 주문서에 따라 "매수인"이 지불한 가격으로 제한된다. 특히 손해배상, 가격 인하, 계약 해지 또는 철회에 대해 명시적으로 언급되지 않은 클레임은 제외된다. 어떤 경우에도 "매수인"은 공급품의 결함을 시정하는데 드는 비용에 대한 보상 외에 손해배상을 청구할 수 없다. 이는 특히 생산 손실, 사용 손실, 주문 손실, 리콜 비용, 이익 손실 및 기타 직간접적 또는 결과적 손해를 의미하지만 이에 국한되지 않는다. 이러한 책임 제한은 공급업체의 중대한 과실 또는 고의적 위법 행위로 인한 책임이나 공급업체의 행위 또는 부작위로 인한 사망 또는 신체 상해에 대해서는 적용되지 않는다.

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Other conditions of sales

1항 : "매수인"은 "매도인"이 직접 또는 "매도인"의 의뢰에 따라 행하는 신용조사에 성실히 응해야 한다.

2항 : 본 계약서에 유첨되는 서류의 기재사항은 본 계약서의 약정사항과 동일한 효력을 가지며 본 계약서와 중복되는 조항은 본 계약서의 조항에 우선한다.

3항 : 본 계약에 명시되지 않은 사항은 일반 상관례에 따른다. 본 계약에 관하여 소송의 필요가 생긴 경우 "매도인"과 "매수인"이 합의하여 관할 법원을 지정할 수 있다. "매수인"과 "매도인"은 본 계약의 당사자로서 신의를 성실하게 이행한다.

4항 : MAS 기계 활성화 시스템 공급자는 계약 합의된 구매 가격의 전액 지불을 고객의 가장 필수적 의무로 간주한다. 따라서 여기에 명시 합의된 지불 조건과 관련하여 전액 지불하지 않거나 그 일부를 지불하지 않는 것은 본 계약의 중대한 위반으로 간주된다. 고객은 장비에 기계 활성화 시스템이 장착되어 있음을 알고 동의한다. 이 시스템을 통해 공급자는 기계 기능의 전부 또는 일부를 일시적 또는 영구적으로 비활성화 할 수 있다. 고객이 지불 불이행 상태에 있는 경우, 계약서에 동의한 대로 시간이 지남에 따라 기계 기능의 전부 또는 일부가 비활성화되어 코드를 입력하지 않는 한 기계가 작동하지 않을 수 있다. 공급자는 고객의 금액 지불로 계약 위반이 해결될 때까지 기계 활성화를 일시 중지할 권리가 있다. 기계 기능의 전체 또는 부분 비활성화는 15 일 전에 기계 화면에 메시지가 표시 되어 코드를 입력하지 않으면 기계가 비활성화 된다는 것을 고객에게 알린다. 이 알림은 기계 기능의 전체 또는 일부가 비활성화되기 전에 적어도 두 번 표시되어 기계가 작동하지 않을 때까지 남은 일수를 고객에게 알린다. 고객이 공급자에게 서면으로 품질 불만을 전달한 경우, 공급자는 합의된 모든 사양을 충족할 때까지 기계가 전체 기능을 유지하도록 보장한다. 공급자는 고객 클레임에 대한 적절한 조사 후 그러한 클레임이 근거가 없다고 합리적으로 결론을 내릴 경우, 기계의 전체 기능 활성화는 연장되지 않으며, 대금 지불 기한은 이후 15 일 또는 최초 만기일 중 더 늦은 날짜로 적용된다. 고객이 관할 법원에 클레임을 제기하지 않는 한 적용된다. 공급자는 이러한 조건에 따라 기계가 자동으로 비활성화될 경우, 비용 손실 또는 손상에 대해 책임지지 않는다. 고객은 시스템을 제거, 비활성화, 수정 또는 변경할 권한이 없다. 이를 위반하면 기계 보증이 무효화된다.

러시아 또는 벨라루스로 재수출 금지 조약

비 EU 국가에 사업장이 등록된 구매자가 EU이사회 규정 제12조 833/2014호 또는 제8조 2024/1865호의 범위에 해당하는 경우 다음 사항이 적용됩니다:

- (1) 구매자는 본 계약과 관련하여 공급된 물품 중 EU이사회 규정 제 833/2014 제12g조 또는 2024/1865 제8g조의 범위에 해당하는 물품을 러시아 연방 또는 벨라루스에 직접 또는 간접적으로 판매, 수출 또는 재수출하거나 러시아 연방 또는 벨로루시에서 사용하기 위해 판매, 수출 또는 재수출 하는것은 불가합니다.
- (2) 구매자는 (1)항의 목적이 리셀러를 포함하여 상업 체인의 하위에 있는 제3자에 의해 좌절되지 않도록 최선의 노력을 기울여야 합니다.
- (3) 구매자는 리셀러를 포함하여 상업 체인의 하위에 있는 제3자가 (1)항의 목적을 좌절시킬 수 있는 행위를 감지하기 위한 적절한 모니터링 메커니즘을 설정하고 유지해야 합니다.
- (4) (1), (2), (3)항의 위반은 본 계약의 필수 요소에 대한 중대한 위반으로 간주되며, 판매자는 다음을 포함하되 이에 국한되지 않는 적절한 구제책을 모색할 권리가 있습니다:
 - (i) 본 계약의 해지.
 - (ii) 본 계약의 총액의 20% 또는 수출된 상품 가격중 높은 금액의 위약금.
- (5) 구매자는 (1), (2), (3)항을 적용하는 데 문제가 있는 경우, (1)항의 목적을 좌절시킬 수 있는 제3자의 관련 활동을 포

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함하여 즉시 판매자에게 알려야 합니다. 구매자는 판매자가 (1), (2) , (3)항에 따른 해당 의무 준수 정보를 요청하는 경우 2주 이내에 판매자에게 제공해야 합니다.

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MACHINE DETAILS



The MIKRON HSM500 HSCI is based on a fixed portal design. The X-axis is on the base of the machine; the Y- axis and Z-axis are mounted on the portal. This design minimizes the total floor space required by the machine and is very well suited for the production of high value components such as molds, electrodes and production parts especially in the range of 3-D.

The portal and the base of the machine are made of polymer concrete. This composite offers excellent vibration damping characteristics - approximately 10 times greater than cast iron and significantly better than welded construction. A high degree of damping is essential when working with high feed rates, high acceleration, and requirements for superior surface finishes. The structure of the MIKRON HSM500 HSCI makes it ideal for difficult applications, such as hard metal machining, where any transmitted vibration leads to increased wear or fracturing of the cutter. The entire machine weighs approximately 5500kg (12125 lbs). Most of the weight is in the base and the portal. The relatively large mass provides for a very rigid machine tool structure. The polymer concrete base has 10 times the dampening characteristics of cast iron. Also polymer concrete is significantly more thermally stable than cast iron.

The worktable, X- axis, Y axis and Z-axis slides are made of high strength cast iron. These elements, reinforced by longitudinal ribs and wide sections, assure that the machine components provide the necessary rigidity. The supporting areas of the guides are ground enhancing the total geometrical accuracy.

- Rugged monoblock portal design with a compact foot print
- Powerful integral spindles with vector control provide high torque at low speeds

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- Tool changer with few moving parts
- High resolution glass scales on all axes
- Excellent access to working area for operator or crane loading
- Heidenhain TNC 640 (Gen.3)
- Polymer concrete construction

The laser tool measuring is positioned in the toolchanger area and separated from the working area by an automatic door. Therefore it is protected against chips and dust.

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INCLUDED FEATURES

MACHINE

995001123 MIKRON HSM 500 - Heidenhain TNC 640

INTELLIGENT SPINDLE MONITORING

This smart machine module displays the Milling process-related vibrations in X, Y and Z directions plus all relevant Spindle temperatures.

With this information, the operator can optimize the process to save costs on Tooling, extend the machine lifetime, increase the part quality and reduce the machining time.

INTELLIGENT THERMAL CONTROL

The intelligent thermal control is a thermal stabilization of the machine, which essentially differs in its operation and precision from all spindle heat compensations systems known until now. Higher precision in shorter time thanks to faster thermal stability of the machine is the main feature. A preheat cycle of the machine is not necessary any more for typical applications, it is only recommended for achieving the highest precision.

High surface quality and geometric precision are now attainable without specific user knowledge of the thermal machine behavior. As the thermal stability of the machine can be influenced by the environment it is necessary for high precision parts to use the machine in a room with a constant temperature (thermally stable room), also the use of coolant liquid may have negative influences.

ANGLE TABLE 550 X 450 MM (21.6" X 17.7")

The well ribbed heavy cast iron angle table is rigidly mounted to give stability and high load carrying capacity. The axial T-slots (parallel to X-axis) facilitate part clamping or optional accessories. No pallet needed.

Clamping surface:	550 x 450 mm (21.6" x 17.7")
T-slots:	1 x 14-H7 (in the center) / 6 x 14-H12
Distance between:	63 mm (2.48")
Max. table load:	200 kg (440 lbs)

ETHERNET INTERFACE

The Ethernet interface is standard in the control. It prepares the control for the connection with an Ethernet network. The communication is via the TCP/IP protocol. All other preparation for the data transfer such as NFS SW program for PC Server and Network installation (Network file system) are the customer's responsibility.

GUIDE SYSTEM

All HSM machines use high quality linear guides. The bearing surfaces of the guides are ground to optimize the straightness and support of each axis and to ensure the overall geometric precision of the machine. The recirculation bearing packs utilize balls for minimal friction and are widely spaced to increase the overall stiffness. The automatic central grease lubrication systems require minimum maintenance, so no need to manually lubricate the guides. In comparison to other guide systems, the linear guides with ball elements provide a number of advantages. Linear guides with ball elements provide higher dynamics and the friction is also significantly reduced, allowing for improved contouring without "stick-slip." When compared to hydrodynamic guide systems, the damping characteristics of linear guides with ball elements are lower. However, the HSM compensates for this by the use of the polymer concrete

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structure that offers superior damping capacity. In spite of the low cutting force, dynamical stiffness is very important to high-speed milling machines because the axes move at very high speeds and are stopped and started and change direction abruptly. The feed axes design is optimized for applications in high-speed milling. The acceleration of all axes is 1 g. The high dynamic drive motors connected to ball screws allow for the highest possible dynamics of the traverse movements and the highest degree of contouring precision.

Control

The control type is Heidenhain TNC 640 (Gen.3)

Feature	Description
Components	Main computer with TNC keyboard and integrated 19-inch. Inter Core i7 processor. Touch screen with soft keys.
Operating system	HEROS 5 operating system for machine control
Memory	RAM memory (MC 8532): 4 GB for control applications. Solid state disk with 21GB free capacity
step	Input resolution and display Linear Axis: to 0.1 μm Angular Axis: to 0.0001°
Interpolation	Linear in 4 Axis including spindle
Block processing time	0.5 ms (3-D straight line without radius compensation)
Error compensation	Linear and nonlinear Axis error, backlash, reversal spikes during circular movements, thermal expansion and static friction
Data interfaces	For use by end users, the MC is equipped with the USB 3.0/ USB 2.0, and Ethernet interfaces. HEIDENHAIN software TNCremoNT or TNCremoPlus 2 x 100BaseT Fast Ethernet interface 2 x USB
Diagnostics	Fast and simple troubleshooting through integrated diagnostic aids

Furthermore this option includes the standard HEIDENHAIN option 8.

Digital Control Design

Uniformly digital

In the uniformly digital control design from Heidenhain, all components are connected to each other via purely digital interfaces: The control components are connected via HSCI (Heidenhain Serial Controller Interface), the new real-time protocol from Heidenhain for Fast Ethernet, and the encoders are connected via EnDat 2.1 or 1Vpp, the bidirectional interface from Heidenhain. This achieves a high degree of availability for the entire system. It can be diagnosed and is immune to noise – from the main computer to the encoder. These outstanding properties of the uniform digital design from Heidenhain guarantee not only very high accuracy and surface quality, but rapid traverse speeds as well.

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HSCI

HSCI, the Heidenhain Serial Controller Interface, connects the main computer, controller(s) and other control components. HSCI is based on 100BaseT Ethernet hardware. A special interface component developed by Heidenhain has shortened cycle times for data transfer.

Main advantages of the control design with HSCI:

Hardware platform for flexible and scalable control system (e.g. decentralized axis systems)

High noise immunity due to digital communication between components

Hardware basis for the implementation of "Functional Safety"

Simpler wiring (initial operation, configuration)

Inverters remain connected via proven PWM interface

Greater cable lengths in the entire system (HSCI up to 70m)

Basic principle

The basis of the HEIDENHAIN safety concept is the dual-channel processor structure, which consists of the main computer (MC) and one or more drive controller modules (CC= control computing unit). All monitoring mechanisms are

designed redundantly in the control systems. Safety-relevant system data are subject to a mutual cyclic data comparison.

The HEIDENHAIN safety concepts enables the setup of a system that fulfills the requirements of Performance Level d, Category 3 in accordance with DIN EN ISO 13849-1 and SIL 2 in accordance with IEC 61508 (DIN EN 61508-1).

It provides safety-related operating modes, as per DIN EN ISO 16090-1 (previously DIN EN 12417). Comprehensive operator protection can thus be implemented. Safety-relevant errors always lead to safe stopping of all drives through defined stop reactions. Defined stop reactions are triggered and safe operating statuses are achieved via safety-relevant inputs and outputs (twin channels), which have an influence on the system in all operating modes.

Peripheral Bus

The System supports Profinet.

Components

In systems with functional safety, certain hardware components assume safety-relevant tasks.

Systems with FS may consist of only those safety-relevant components which have been enabled by Heidenhain, including their variants!

Control components with functional safety are recognizable by the suffix FS after the model designation, e.g. MB 735 FS.

MB and TE

An MB machine operating panel with FS is indispensable for Systems with functional safety. Only in this MB do all keys have a dual-channel design.

PLB

In systems with functional safety, a combination of hardware (FS and standard) is possible, but a PLB 62xx FS is essential.

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HR

FS handwheels are required in systems with functional safety because only they have the required cross-circuit proof permissive buttons.

By the FS handwheels the customer need only press one permission button, because each button is a dual channel button.

Safety functions

The software process of the safety-kernel software (SKERN) and the SPLC run in parallel on the MC and CC. Basic safety functions are permanently defined in the SKERN software.

The safety-kernel software receives status information and requests for safety functions from the SPLC. The SKERN triggers safety functions and monitors them. Furthermore, all dynamic tests are controlled by the safety-kernel software.

The safety-kernel software is responsible for the realization of all basic safety functions:

Triggering and monitoring of the stop reactions (SS0, SS1, SS1F, SS2)

Standstill monitoring in SOS state

Monitoring of the safely limited speeds (SLS) in the various safety-related operating modes

Safely-limited position (SLP)

Nominal-actual value comparison of position values or speed values

Control of dynamic tests

Carrying out the cross comparison

Transfer of axis-group states (STO, SOS, AUTO (AUTO = operation if the guard doors are closed) or of the safety function in direct connection with the operating mode: e.g. SLS_2 through SLS_4) to the SPLC

Transfer of the axis states (at standstill or in motion) to the SPLC

Transfer of the axis positions to the SPLC

Performing the safety self-test (SSt)

QUALITY PARAMETERS - HSM500

The HSM500 high-speed milling machines are constructed and tested according to certain quality standards. All three axes comply with positioning tolerances required by the ISO 230-2 standards:

Positioning accuracy - X-axis (A):	8 µm (0.00031")
Positioning accuracy - Y-axis (A):	8 µm (0.00031")
Positioning accuracy - Z-axis (A):	8 µm (0.00031")
Deviation of position - X-axis (M):	5 µm (0.00019")
Deviation of position - Y-axis (M):	5 µm (0.00019")
Deviation of position - Z-axis (M):	5 µm (0.00019")
Repeatability plus - X-axis (R+):	5 µm (0.00019")
Repeatability plus - Y-axis (R+):	5 µm (0.00019")
Repeatability plus - Z-axis (R+):	5 µm (0.00019")
Repeatability minus - X-axis (R-):	5 µm (0.00019")
Repeatability minus - Y-axis (R-):	5 µm (0.00019")
Repeatability minus - Z-axis (R-):	5 µm (0.00019")

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Backlash - X-axis (B):	4 µm (0.00015")
Backlash - Y-axis (B):	4 µm (0.00015")
Backlash - Z-axis (B):	4 µm (0.00015")

To ensure the highest quality results, the machine must be installed at an appropriate location. A constant room temperature is of particular importance. The positioning accuracy is guaranteed at a temperature of 22°C ±1°C (71.6°F ±2°F). The machine must also be positioned to avoid direct sunlight and drafts.

SAFETY DEVICES

The door is equipped with an interlock to prevent operation of the machine when the door is open. All HSM complies with CE regulations.

CONNECTIVITY

DNC Option 18 is set by default. This option enables machine communication and allows publication of internal machine information and status.

600.50/01 Machine with EC conformity declaration

With this option, the ordered machine configuration is built and delivered in conformity with 2006/42/EC.

205.10/01 Machine encl. polycarb. window

The protection cabin closes the machining room. The circular slide door can be opened in one single movement in front, sideways and at the top. In the side wall a trapezoid window is installed. The loading by crane is possible. Thanks to the integrated lighting the machining room is perfectly illuminated. The ergonomics of the machine is designed for the operating comfort, the optimal access to the machining room, and the supervision of the machining process by the operator.

For operational safety reasons, polycarbonate screens must be replaced at least every 2 years. Any collision with foreign objects means that they must be replaced immediately.

205.08/03 Spindle 42000 min⁻¹/ HSK-E40

The cartridge type high frequency spindle is flanged mounted directly into the Z-axis-slide to minimize thermal distortion and provide ease of maintenance. The spindle taper accepts tool holders to HSK standard with retention by a hydro-mechanical system. During tool changing the taper cleanliness is maintained by a continual air blast. The high speed spindle utilizes top class hybrid ceramic bearings lubricated by a controlled oil-air system and protected against contamination by an air barrier. The complete spindle is temperature controlled by a controlled spindle refrigerating unit by a closed loop water cooling system. This cooling unit has a refrigerating capacity of 4.5 kW (6.04 Hp) and thus allows the machine to be kept constantly refrigerated (motor spindle and electric cabinet) even in fluctuating temperature environment. The controlled spindle refrigerating unit is placed by the side of the machine. Monitoring systems and easy to read media state indicators are standard features.

Datas of the vector driven high frequency spindle:

The following specifications are valid for solid carbide end mills in shrink-fit toolholders, balanced at G2.5 for HSK-E as per standard ISO 1940 by full speed.

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Power at S1	10 kW
Max. Torque at S1	3.2 N
Power at S6 (40% ED, ts 2min)	13 kW
Max. Torque at S6 (40% ED, ts 2min)	4.2 Nm
Acceleration and braking time	2.5 s
Speed range, standard	100 - 42k rpm
Ballbearing diameter	45 mm (1.77")
Toolholder type	HSK-E40
Hold- and ejection force	6'800 N (1530 lbf)
Weight Tool and Toolholder	max. 1 kg
Max. Tool diameter	16 mm (0.63")
Max. Tool length	120 mm (4.72")
Vector drive provides max. torque at low speed. Rigid tapping is possible.	

205.09/03 Tool magazine 36× HSK-E40

All the tool changers are built according to the same concept, a disk magazine with a choice of different number of tool positions. The exchange of a tool in to the vertical spindle, takes place by the positioning axis Y and Z. Sensors check the location and position of the tool holder in the magazine. The tools are held with the cutting edges down in the disc-type

magazine, to avoid any chips or coolant reaching the tool taper.

The tool changer is separated from the working area by a stainless sheet metal with an integrated automatic door. Therefore the tools are protected against chips and dust during machining. The magazine is easily accessible from the side of the machine and can be loaded during machining.

Tool capacity:	36
Tool interface:	HSK-E40
Max. tool diam. external row HSM 500:	65 mm (2.56")
Max. tool length external row HSM 500:	200 mm (7.87")
Max. tool diam. external row HSM 700:	65 mm (2.56")
Max. tool length external row HSM 700:	210 mm (8.27")
Tool-to-tool time:	10s
Chip-to-chip time (VDI 2852):	12.5s

The maximum tool diameter and length is only for touch probes.

600.12/58 Robot interface

This sales item enables the operation of a GF Profinet Robot Interface homologized robot system on the machine tool. It consists of an automatically actuated door as well as an electrical connector plug. The valves for the chucks and clamping units are always on the machine. The robot controls the chucks and clamping units by electric signals.

This sales item creates a separate access to the workspace for the automatic pallet change. This in addition to the main operation door.

The software option DNC 500.52/13 is a standard option on all Heidenhain machines. DNC is mandatory for the communication between the milling machine and a cell management system, such as System 3R WSM.

For the Sinumerik-One the OPC-UA is mandatory for a Job Management System.

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By a Fanuc machine is Focas always available.

The supported couplers are the

- ANYBUS coupler for System 3R
- B&R for EROWA and
- the Siemens 6ES7158-3AD10-0XA0 PN/PN Coupler for OEMs.
- Other couplers must be requested by an IMM

Please note:

- VCE Pro series are not GF Profinet but GF AC-3PAI interface.
- The CE conformity for the overall system is guaranteed by the general contractor.
- In cases the general contractor is a third party, the machine is handed over to this third party with a Manufacturer's Declaration.
- Safe commissioning in the automation system is the responsibility of the general contractor.
- The robot system has to be 100% compatible and homologized with the GF Profinet Robot interface.

Important note related to the Mikron MILL P 500 U:

Mikron MILL P 500 U as part of an automated cell is not protected against access from the rear.

For operations within the protected area of the automated cell or when performing maintenance operations of the automated cell, GFMS recommends to shut down the machine at all times.

If this is not possible, the automation integrator of the cell must make provisions for the safe operation mode of the machine by the integration of appropriate safety devices (such as light curtains or alike) at the interface to the rear side of the machine. It is the operator's responsibility to set up such safety devices, to check for functionality and to enforce the appropriate use of it.

GFMS strongly recommends to review any installation and the according working instructions with experienced automation integration contractors.

Differing from the standard interface, a special request has to be placed for any kind of customized Robot systems on machine tools.

This specifically applies a combination of palette and part handling.

The requirement specifications for the system will be worked out in close cooperation between the different parties in an early stage. A quote based on the requirement specification will be worked out including price, delivery time and responsibilities for the customized robot interface.

For each machine with a Robot Interface, the Automation Questionnaire must be completed.

CHIP DISPOSITION

205.11/01 Coolant tank

The coolant tank serves to take the coolant. The chips fall into the tray underneath the machining room. Manual disposal of the chips. Spray gun and coolant supply to work zone included.

Capacity: 150 l

Pump pressure: 3 bar

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Pump capacity: 30 l/min

COOLANT & DUST EXTRACTION

292.12/13 Mist extraction unit

The soiled air is aspirated through two preliminary filters and purified from coarse dust and liquid particles of any kind. The remaining particles are electrically loaded by ionization and are then collected on the alternatively loaded or earthed metal plates of the electric filter cells. The system is completely integrated in the enclosure. The system requires regular maintenance.

Note: Not built for heavy coolant mist

292.12/20 Air blow off external

Air blow off nozzle can cool the tool and part and blows off the chips. It can also blow dust up and make it easy for the extraction unit to suck the dust out when machining the graphite.

OPTIONS

600.12/19 Beacon

The beacon allows to monitor the operating status of the machine from a distance.

The operating status is indicated by three signal colours.

Green: Program in progress

Red: Program stopped or complete

Flashing red: Fault

PRECISION & ACCURACY

600.12/13 Laser measuring system

The laser measuring system is used for tool measurement, wear and breakage control. The tool length can be measured at the spindle speed relevant for machining. The measuring cycles for length and radius measurement, temperature offset, calibration, individual cutting edge and tool breakage check can be set in the control system. It is also possible to check eccentric tools and roughing cutters. The optics of the laser are protected by a cap and with overpressure.

To improve the reliability of the measuring process, we recommend the tool blow-off unit 600.12/20.

Manufacturer	BLUM Novotec
Product	Mikro Compact NT 200 mm
Laser safety class	Class 2 in accordance with IEC825 21 CFR 1040.10
Laser type	Visible red-light laser / 630 – 700 nm/ < 1 mW
Type of protection	IP68
Current supply	24 V DC / 160 mA
Repetition accuracy of the laser	+/- 0.001 mm

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Minimum tool diameter according to manufacturer's specifications	0.1 mm
Minimum tool diameter use on the machine with process reliability	0.6 mm
Test speed	750 – 150.000 U/min

Note: The measuring accuracy can vary on the machine due to the operational conditions.

Note: Machine-specific restrictions on the measurable tool geometry are listed in the specification section. The position of the laser is also listed in the specification section.

600.12/20 Tool blow-off unit

To improve measuring reliability when using a laser, the blow-off unit can be used to blow-off light deposits from the tool tip before the measurement takes place. The device is attached at the side of the laser and is automatically activated by the measuring cycle.

600.12/08 Touch probe infrared HSK-E40

This touch probe is suitable for workpiece setup on the machine as well as to check the work results. Based on optical signal transmission, the touch probe can be stored in the tool magazine and changed into the spindle, either automatically or manually.

Manufacturer	M&H
Type	40.02 -HSK 40-SM
Housing diameter	40 mm
Housing length	50 mm
Protection category	IP68: EN60529 (15 m) tested

Repetition accuracy of the touch probe in one direction (max. 2 Sigma)	max 0.5 µm(2 Sigma) with 50 mm stylus and 254 mm/min feedrate
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Recommended feed rate	Max. 2000 mm/min
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Measuring directions	±X, ±Y, -Z
Overtravel of the probe insert	XY ± 12.5°; Z -5 mm
Trigger force with 50mm stylus	XY = 0.8 N, Z = 5.7 N

Battery type	2 x Battery(3.6V 1/2AA) Standby 12 months
Mass without Shank	approx..235g
Temperature range	Operation:5°C – 55 °C, Storage: 5°C – 70 °C
Material	Stainless steel
Maximum Probing Frequency	50Hz= 50 points/s

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Note: The measuring accuracy can differ in practice due to the operational conditions.

600.12/10 Preparation touch probe infrared

This option equips the machine with an infrared signal transmitter/receiver inside the working area. Thus the machine control system is capable of communicating with a compatible infrared touch probe.

OMI-2 differs from the OMI in that it offers the new modulated transmission method, providing increased resistance to optical interference.

SMART MACHINE

500.52/01 APS Advanced Process System

The idea of APS is to give the operator a tool to supervise and optimize the milling process. Critical milling strategies can be recognized and optimized. The advantage for production applications is the possibility to optimize the milling process without damaging the motor spindle or the machine.

Further advantages are improvement of work pieces surface finishes, higher tool lifetime as well as lifespan increase of motorspindle and machine.

500.52/07 ITC Intelligent Thermal Control

The intelligent thermal control is a thermal stabilization of the machine, which essentially differs in its operation and precision from all spindle heat compensations systems known until now. Higher precision in shorter time thanks to faster thermal stability of the machine is the main feature. A preheat cycle of the machine is not necessary any more for typical applications, it is only recommended for achieving the highest precision.

High surface quality and geometric precision are now attainable without specific user knowledge of the thermal machine behavior.

As the thermal stability of the machine can be influenced by the environment it is necessary for high precision parts to operate the machine in a room with a constant temperature. Also the use of coolant may have negative influences on thermal machine behavior.

500.52/19 OSS Operator Support System

OSS Operator Support System optimizes the machining process according to the requirements of the workpiece. With cycle 311 it is possible to specifically define the target parameters for speed, accuracy and surface quality and to individually modify the workpiece weight at any time. The operator can use the OSS to configure the machine settings. The four predefined settings can be selected. These settings can be activated manually or in the NC program.

Customer benefits

- The intelligent system precisely adjusts the dynamic behavior of the machine for the requirements of each workpiece
- Shorter machining time
- Improved surface quality
- Better accuracy
- Reduced difficulty for workpiece retraction

Machining priorities:

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Time

During the roughing process, time is of greatest importance. When the user selects time as the highest priority, the OSS extends the tolerance band. In addition, the control system adapts the speed profile to the geometry to be processed.

Accuracy

If the workpiece requires maximum precision, a narrower tolerance band must be selected to attain these levels of accuracy. The control system is configured so that the best possible geometric precision is guaranteed.

Surface quality

Attaining optimal surface quality despite a poor quality of NC program requires enlarging the tolerance band and smoothing the transitions in the tool path.

Weight classes

The optimal settings for the drive are based on the specifications for the workpiece weight. Sliding friction and static friction, which increase based on weight, are compensated for.

Technical specifications

Machining priorities: 0 Standard, 1 Surface Quality, 2 Time, 3 Accuracy,

Weight classes: 0 light, 1 medium, 2 heavy

Activation: manual or with an NC program

500.52/13 DNC Remote network access on control

Communicate with an external PC over the network. This option is necessary for connecting the Mikron Mill machine to a cell management system, such as System 3R WorkShopManager or EROWA JMS. The DNC option is already integrated into the optional robot interface attachment or System 3R WorkShopManager.

PACKING & DOCUMENTATION

600.21/02 Packing HSM 500 Type C

Standard packing for ocean freight. This packing includes retaining and anchoring materials and the preservation required for seaworthy transport in the form of a wash-off coating for exposed steel and cast parts, desiccant and airtight heat-sealed wrapping in a stable seaworthy wooden box. Thus this packing is suitable for ship transport overseas or transport that results in lengthy interim storage (maximum of 6 months).

600.21/80 Preparation packing Type C

Preparation for standard packing for ocean freight. This packing preparation is intended for external packing and transport carried out by the respective sales company. This preparation includes retaining and anchoring materials and the preservation required for seaworthy transport in the form of a wash-off coating for exposed steel and cast parts, desiccant and airtight heat-sealed wrapping. Thus this packing is suitable for ship transport overseas or transport that results in lengthy interim storage (maximum of 6 months). The wooden platform included in the delivery to the external packing service provider must be returned to the factory within 30 days, as it is not included in the price. If the wooden platform is not returned in a timely manner, the sales company will be charged.

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Contract type embedded with the machine

CSP995011 Bronze - Recover fast

Bronze is the way to get faster assistance through My rConnect.

You are guaranteed 1 hour callback on support requests, 24/5 access to expert technicians, and fast track service intervention scheduling to ensure quick diagnostics, uninterrupted production and efficient troubleshooting.

BRONZE

CSP000024 Yearly base contract

CSP000008 Fast track for support

Fast response time through My rConnect for digital service requests within 60 minutes.

Fast service engineer dispatching for onsite interventions.

CSP000012 24/5 expert support through My rConnect

Remote support through My rConnect can be reached 24 hours a day, 5 days a week (Monday to Friday)

Other service contracts

200011529 My rConnect - Remote Access - Trial 1st year

Remotely access to your machine anytime, anywhere, to monitor its status and optimize your workflow via the My rConnect web platform. Additionally, you can authorize GF Machining Solutions technical support to troubleshoot your machine remotely. This secure service is available through an annual subscription.

200011531 My rConnect - Live Data - Trial 1st year

Stay informed about your machine status anytime, anywhere. Monitor machine status and operation progress, receive notifications for status changes, messages, or alarms, and react quickly without needing to visit your plant. Combined with Remote Access, you can also view your machine's user interface remotely. This feature is available through an annual subscription.

THIRD PARTY PRODUCTS

Z100002 Air Dryer

Z100002 Stabilizer

SERVICES

Z100003 Basic machine Installation

Z100003 Operation Training

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SPECIFICATIONS

Machine total weight	5500 kg (12'125 lbs)
X Axis Travel	500 mm (19.685")
Y Axis Travel	450 mm (17.73")
Z Axis Travel	350 mm (13.79")
Max. Feed force (X)	8.35 kN (1878.75 lbf)
Max. Feed force (Y)	5.15 kN (1158.75 lbf)
Max. Feed force (Z)	4 kN (900 lbf)
Recommended max. feed rate	42 m/min (1654 in/min)
Rapid traverse (all axes)	42 m/min (1654 in/min)
Maximum acceleration per axis	1 g
Table surface	550 mm x 450 mm (21.65" x 17.72")
Base	Polymer Concrete
Measurement system resolution 1024 increments	0.021µm 0.0000000827 in
Max. table load	200 kg (441 lbs.)
Control	Heidenhain iTNC530 HSCI FS
Noise Emissions	71 dB(A)

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ENVIRONMENT AND SERVICE REQUIREMENTS

Electrical

Main voltage:	3 phase 400 Volt 50/60 Hz
Permitted main frequency fluctuation:	$\pm 2\%$
Total installed rated load:	25 kW
Permitted main voltage fluctuation:	$\pm 6\%$ at 3 phase 400 Volt at no load and full load
Control voltage:	24 Volt DC

Compressed Air

The compressed air temperature must correspond with the ambient temperature of the machine location to assure a high degree of machining precision. The standard pneumatic unit includes a fine filter, an automatic steam trap, an automatic drain, as well as valves and a pressure gauge. An On/Off valve connected to the main switch will prevent unnecessary consumption of compressed air when the machine is turned off.

Min. pressure:	6 bar (87 psi)
Min. quantity:	Approx. 100 l/min (26.4 gal/min)
Compressed air quality:	5.4.5 according to ISO 8573.1

Foundation Requirements

With regard to service specifications related to high work piece requirements, the machine has to be installed in appropriate surroundings. The floor has to be flat and free of disturbing influences, such as vibrations.

Network Requirements

An input voltage tolerance of $\pm 6\%$ must be available. If fluctuation is $> \pm 6\%$, a voltage stabilizer must be installed. Lower voltages can damage the electronic components and will void the warranty. Conditioned power lines are recommended if the location is susceptible to power spikes and brown-outs or the power is of poor quality. For additional information, contact the machine manufacturer.

Protection of the supply line:	Typical time lag 40A, not surge protected
Cable cross section at UN 400V:	10 mm ² /AWG #8

Temperature

(IEC 60204-1, Chapter 4.4.3/4.4.4/4.4.5)

Safe ambient temperature:	+10°C to +35°C (+50°F to +95°F) Equipment operation
Safe ambient temperature:	+22°C $\pm 1^\circ\text{C}$ (+71.6°F $\pm 2^\circ\text{F}$) Positioning precision
Safe ambient temperature:	-30°C to +55°C (-1.2°F to +131°F) Stocking + transport
Max. Safe relative humidity:	75% air temperature class H

My rConnect

Digital services for all your GF machines, anytime, anywhere

My rConnect is your gateway to advanced digital assistance. This innovative web-based platform provides seamless access to a wide range of digital services to boost machine uptime and productivity. Enjoy simple navigation and direct access to GF support, for all connected and unconnected machines, regardless of their age.

Comprehensive and free-to-use asset management tool for all GF machines

 My rConnect



Organize your workshop and facilities, and manage user access



Use chat, video conferencing, whiteboard, and file transfer for rapid interaction and problem-solving



Manage and access the complete history of service cases for improved decision-making



Easily access machine user manuals in one place, and upload additional documentation

My rConnect

Connect your machine for greater responsiveness and operational efficiency

Get first-class support and real-time insight by subscribing to Remote Access and Live Data service offerings, available on-demand for eligible machines.

Do you want to increase your equipment uptime?



Remote Access

- Connect and share files to/from your machine from anywhere to solve issues efficiently
- Get expert troubleshooting from GF with remote access to your machine's HMI

Do you know if your machine is operating now?



Live Data

- Increase your responsiveness to unexpected events with real-time status notifications and instant alerts.
- Track machine connection and job status anywhere, anytime

Exclusive support when you need it most

Combining Remote Access and Live Data with Success Packs enhances responsiveness and optimizes uptime.

For more information, contact your local Sales Representative.

Bronze

Silver

Gold